Reverse-Engineering Mathematical Reasoning from OpenAI's 2026 Mathematics Corpus: Implications for Mechanistic Interpretability

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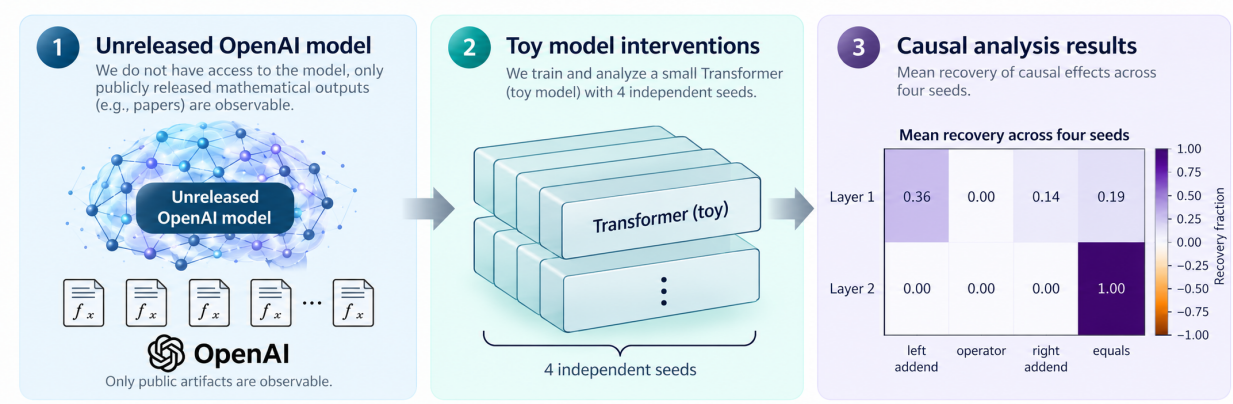


Figure 1: Overall conceptual workflow. The public OpenAI mathematics release supports an artifact audit but does not expose model internals. We therefore conduct causal residual-stream interventions only in separately trained, inspectable toy Transformers, then summarize recovery across four independent initialization seeds.

Abstract

OpenAI's public 2026 mathematics release is an unusually rich public record: mathematical manuscripts, source artifacts, Lean materials, and ten abridged reasoning summaries. It is not a release of model weights, activations, full prompts, or complete internal reasoning. We present a reproducible, preregistered artifact audit at a pinned release commit. We find 372 result families and 719 catalogue-linked manuscripts, with every linked PDF and TeX-source directory present in the Git tree. We identify 242 families with a Lean scope document and 10 released abridged summaries. Catalogue-linked manuscript count does not meaningfully differ by scope-document availability (Mann–Whitney $p = 0.638$; permutation $p = 0.665$; rank-biserial $r = 0.03$). In a separately specified exploratory extension, we structurally index every versioned Git blob (133,955 blobs; 2.06 GiB) by path, object identifier, size, and artifact role. In a Version 2 exploratory continuation, we further extract 45,620 literal TeX environments and 97,343 within-document source references from 8,007 source files, while separately retaining 42,143 rule-based motif candidates. These structures are not proof verification or semantic annotation. These observations do not reverse-engineer neural mechanisms. To make that boundary operational, we separately conduct a causal residual-stream patching simulation in 4 fully inspectable toy Transformers. That surrogate result demonstrates an intervention method with seed-level uncertainty, not a property of an OpenAI model. We release code, provenance hashes, derived data, figures, and an arXiv-ready source package.

1 Introduction

Reasoning traces expose intermediate claims, proof organization, and references to verification. Yet a persuasive chain of thought is not automatically a faithful account of the internal computation that produced an answer [14, 30]. Mechanistic interpretability requires internal access and causal intervention, as in work on sparse features and circuit-level representations [8, 29].

OpenAI's October 2026 public mathematics repository contains manuscripts and supporting artifacts produced by an unreleased internal model [26, 27]. It is valuable for studying a public record of model-assisted mathematics, but it should not be read as a transparent window into neural computation. We provide a pinned inventory, a preregistered artifact-level comparison, and descriptive measurements of the published summaries. We then separately demonstrate a genuine causal mechanistic-interpretability intervention in a toy model.

2 Scientific context: mechanistic interpretability

Mechanistic interpretability seeks explanations expressed in a model's actual computations: features, states, edges, and algorithms that can be interrogated and causally manipulated. The circuit perspective developed from feature visualization and transformer-circuit analyses [7, 23, 24] is deliberately stronger than a useful behavioral description. It asks whether a proposed mechanism predicts the effects of intervention.

There are instructive examples in accessible models. Work on induction heads, modular-addition grokking, indirect-object identification, and a GPT-2 greater-than capability combines structure, activations, and interventions to propose circuit-level accounts [13, 22, 25, 32]. Those successes are neither a license to infer circuits from prose nor a guarantee that a localized explanation is complete. They instead specify the kind of evidence missing for an unreleased model.

Causal tools make the distinction concrete. Causal mediation, activation and path patching, model editing, automated circuit discovery, and causal scrubbing each test particular counterfactual claims [4, 5, 10, 11, 20, 31, 34]. They also expose failure modes: an ablated computation can be compensated by others, and a single intervention can overstate a component's explanatory role [2, 9, 19].

Recent benchmarks sharpen the evaluation standard. KisMATH aligns extracted reasoning graphs with interventions on mathematical tasks, while MIB compares mechanistic localization methods against shared causal targets [21, 28]. These works rule out a priority claim for graph-informed mathematical interventions here. Instead, the Version 2 continuation asks whether public research-source structure can generate family-held-out causal hypotheses in a separately specified inspectable model.

Feature-level approaches address the fact that individual neurons can mix several concepts. Toy superposition models, dictionary learning, and sparse autoencoders supply hypotheses about more useful feature bases [3, 6, 8, 29]. In parallel, chain-of-thought and process-supervision research clarifies why visible reasoning can aid evaluation while remaining distinct from the latent computation that determined an answer [14–16, 30, 33]. Our simulation adopts the causal-intervention side of this program while making no claim to have recovered an arithmetic circuit.

3 Evidence boundary

The release contains public outputs and supporting artifacts. It does not release weights, activations, attention patterns, optimizer state, complete prompts, training data, or complete hidden reason-

ing. Therefore, our strongest permissible claims concern availability, organization, and surface-text statistics. We cannot identify a circuit, feature, attention head, latent representation, or causal sequence in the OpenAI model. Nor can a manuscript or a formalization scope page establish the correctness of a mathematical claim. This distinction matters because process supervision can make intermediate steps useful targets without making prose a complete causal explanation [15].

The title's implication for mechanistic interpretability is methodological. The public record can motivate frozen benchmarks and hypotheses for inspectable models, such as whether a proof segment is causally necessary for a conclusion. It cannot answer that question without a model whose internal states can be recorded and manipulated.

4 Corpus and method

4.1 Frozen source

We acquired the Apache-2.0 licensed `openai/math` repository at commit `fd4aeeb2...0529eeeb`; the acquisition manifest records the full 40-character identifier. The manifest records SHA-256 hashes of `README.md`, `CONTENTS.md`, and `LICENSE`, plus a successful `git fsck`. Windows path limits prevent materialization of a few extremely deep upstream paths in a normal working tree. Rather than silently lose them, the pipeline reads immutable Git blobs directly from the verified object database.

The pre-analysis contract is in `RESEARCH_PROTOCOL.md` and `PREREGISTRATION.md`. The unit is a result family for catalogue measures and a manuscript for file availability. Families are not independent model runs.

4.2 Measurements

We parse each numbered family and each linked PDF in `CONTENTS.md`. For each manuscript, we measure presence of the linked PDF, a non-vendor TeX file, and a README marker. For each family, we measure whether `lean/docs/NNN.md` exists. This is Lean scope-document availability, not verified correctness or successful compilation. For the ten summary PDFs named in the upstream README, we extract pages, word-like tokens, and transparent keyword counts. Failures remain in a warnings table.

The sole preregistered comparison asks whether catalogue-linked manuscript count differs between families with and without a Lean scope document. We report medians, a two-sided Mann–Whitney test, rank-biserial correlation, and a two-sided seed-20261009 label-permutation calculation using 10,000 draws. The latter is a reference calculation, not a claim of independent random families.

4.3 Exhaustive release-tree extension

The catalogue measures do not by themselves cover every versioned artifact. We therefore registered a separate exploratory protocol before executing a structural census of every Git blob at the same immutable commit (`FULL_TREE_INVENTORY_PROTOCOL.md`). The implementation resolves the pinned commit once, reads immutable Git objects rather than the working tree, and records each blob's path, object identifier, byte size, top-level directory, extension, deterministic path/extension-based artifact role, and catalogue or released-summary relation. This is exhaustive at the release-tree level: it includes uncatalogued preprint bundles, sources, configuration, and release metadata rather than treating the catalogue as the denominator. It is not a semantic reading of every file,

a proof check, Lean compilation, or an expansion to all research artifacts OpenAI may publish elsewhere.

4.4 Version 2 exploratory source-structure extraction

Before any Version 2 extraction, we froze a continuation protocol and a separate inactive confirmatory plan. The extractor reads immutable Git blobs for all preprint TeX sources. It records TeX theorem-, lemma-, proposition-, corollary-, definition-, remark-, and proof-environment candidates with source commit, blob identifier, document identifier, result-family relation, and line span. Graph A retains only literal within-document TeX references. Graph B contains fixed lexical rules for candidates such as contradiction, induction, construction, reduction, invariant, case splitting, limiting argument, and equivalence. Graph A is not an entailment graph; Graph B is not semantic or human annotation.

This checkpoint deliberately does not create validated mathematical counterfactuals or execute a new model intervention. Lean kernel verification does not by itself certify informal semantic fidelity [1]; without a detected Lean toolchain and independent semantic audit records, no Version 2 counterfactual is labelled verified.

5 Results

5.1 Release inventory

Artifact	Count
Result families	372
Catalogue-linked manuscripts	719
Families with Lean document	242
Linked manuscript PDFs present	719
Manuscripts with TeX source	719
Released abridged summaries	10

The parser recovered 372 families and 719 linked manuscripts, matching the upstream catalogue headline. Every linked PDF and manuscript directory with TeX sources was present in the pinned Git tree (Figure 3). There are 242 families with a Lean scope document (65.1%). OpenAI separately reports approximately 42% of top-line results as formalized [27]; this uses a different unit and is not interchangeable with our scope-document measure.

5.2 Exhaustive release-tree census

The same release tree contains 133,955 versioned blobs totaling 2.06 GiB. The structural census includes 122,876 Lean source files, 8,007 preprint TeX sources, and 776 preprint PDFs (Table 1; Figure 2). These are file counts and path-based roles, not claims that the Lean files compiled, the manuscripts are correct, or each source file was semantically interpreted.

Exhaustive tree inventory	Count
All Git blobs	133,955
Lean source files	122,876
Preprint TeX source files	8,007
Preprint PDFs	776
Named released summaries	10

Table 1: Exhaustive structural inventory of every versioned blob in the pinned release. Categories are deterministic path/extension roles and their units are files, not mathematical results.

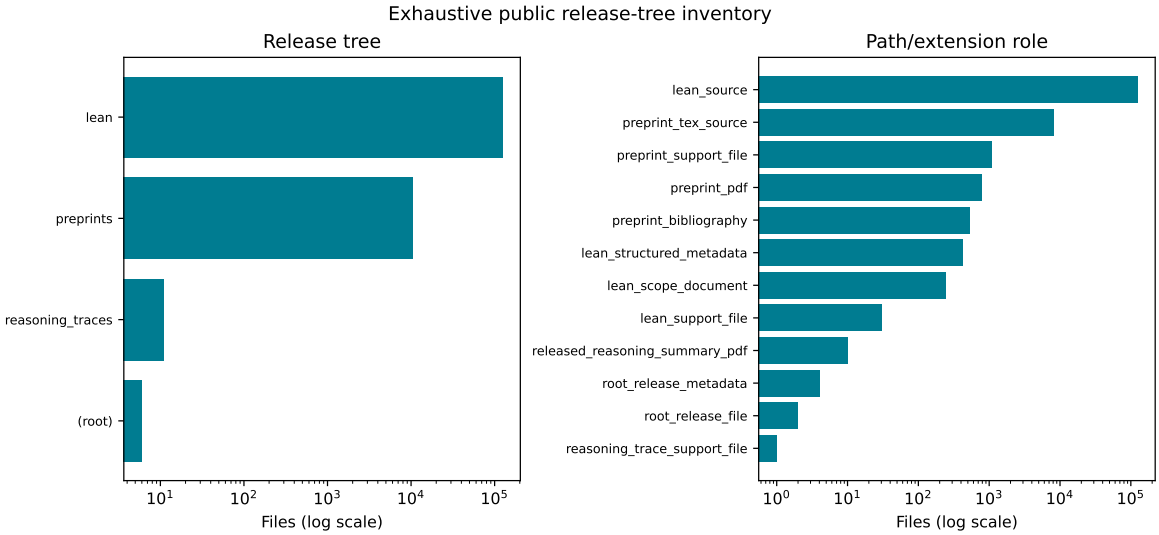


Figure 2: Complete path/extension-based inventory of the pinned release tree. The log scale makes the small root and reasoning-summary groups visible beside the large Lean and preprint collections. This is a structural census of all versioned blobs, not a quality or correctness assessment.

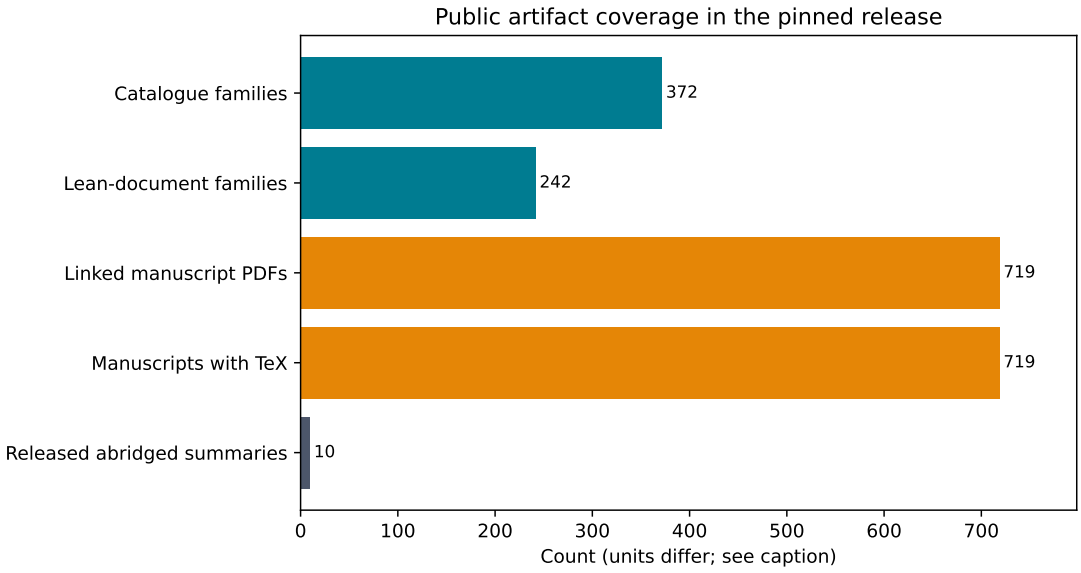


Figure 3: Public artifact coverage in the pinned release. Bars mix family, manuscript, and summary units, so they describe availability rather than a common rate or quality score.

Family size is concentrated at one linked manuscript, with a smaller tail of families containing companion arguments or consequences (Figure 4). This is a catalogue-structure observation, not a count of independent solution attempts.

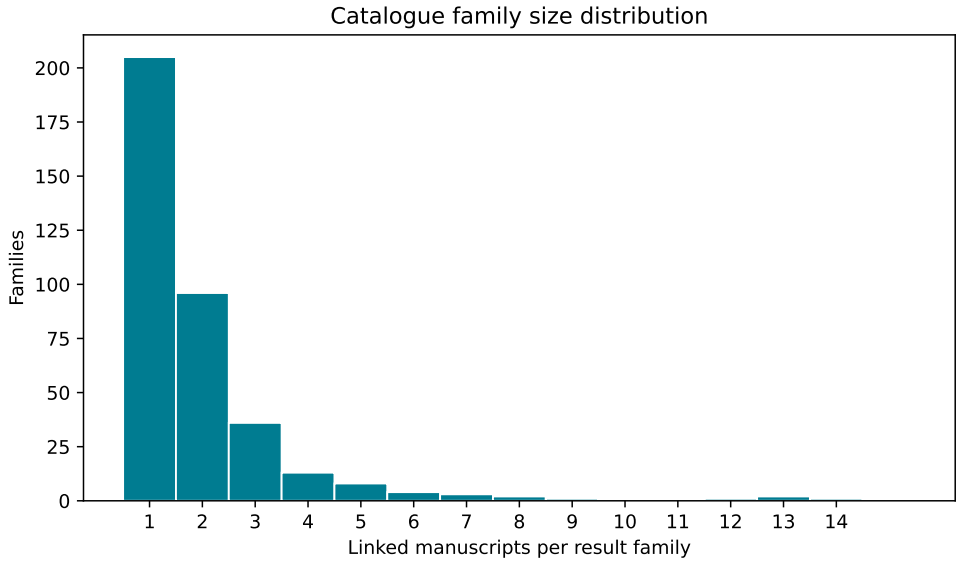


Figure 4: Distribution of catalogue-linked manuscripts per result family.

5.3 Scope documents and family size

Both groups have a median of one linked manuscript. The comparison is small and non-significant ($U = 16{,}150$, $p = 0.638$, $r = 0.03$); the deterministic permutation calculation gives $p = 0.665$. This does not demonstrate equality, causal independence, mathematical quality, or Lean compilation

success. It only gives no meaningful evidence that this simple catalogue-size attribute separates families by scope-document availability at this release.

5.4 Released summaries

The 10 named summaries span 5–45 PDF pages and 2,020–19,935 extracted word-like tokens (Figure 6). Their non-random selection and heterogeneity make them unsuitable for estimating properties of all model reasoning. They can support qualitative study of the published text but cannot recover omitted deliberation or establish faithfulness to unobserved activations.

5.5 Exploratory proof-structure candidates

The Version 2 extractor processed 8,007 TeX files, yielding 45,620 source-environment candidates, 97,343 deduplicated literal reference edges, and 42,143 lexical motif records. It could not resolve 15,824 labels within their source document; those references are excluded rather than guessed. Figure 5 is a frequency display of rule matches, not a prevalence estimate for proof strategies. Repeated document templates, prose conventions, and unequal source lengths may all influence these counts.

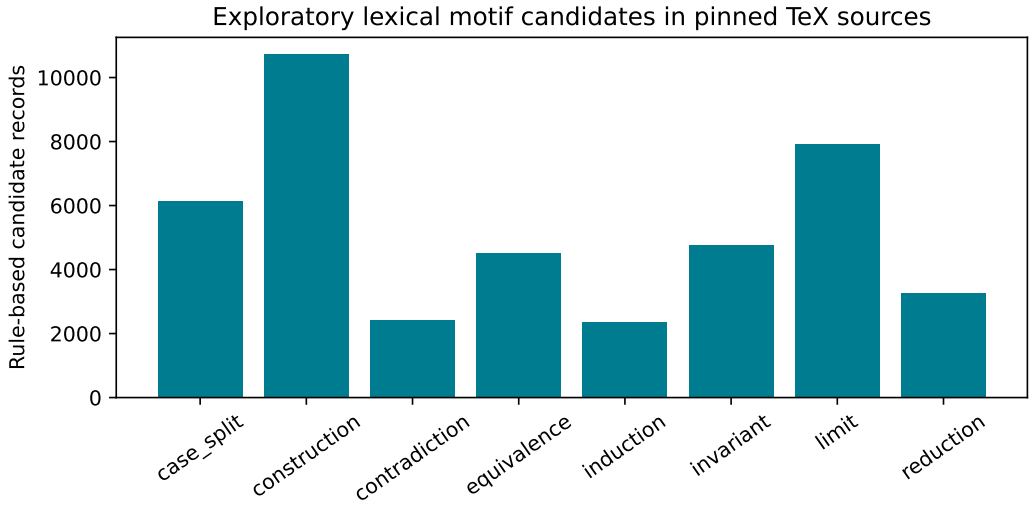


Figure 5: Exploratory lexical proof-motif candidates from pinned TeX sources. Each bar counts a fixed regular-expression match in a candidate environment; it is neither a semantic annotation nor evidence of a mathematical dependency.

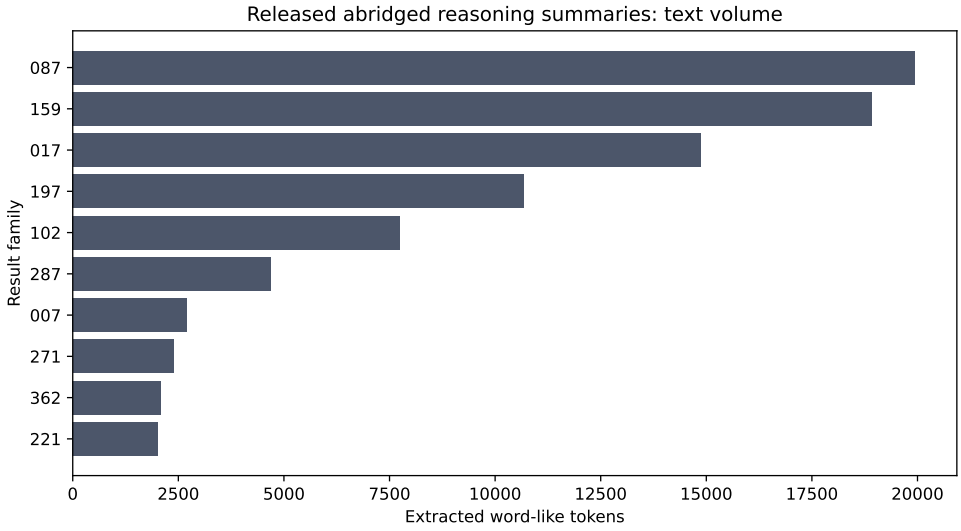


Figure 6: Extracted word-like token volume of the ten explicitly released abridged summaries. This is descriptive parser output, not a reasoning-quality or faithfulness metric.

6 Causal intervention in an inspectable surrogate

To distinguish output analysis from a mechanistic experiment, we preregistered the extension in `SIMULATION_PROTOCOL.md` before running 4 independently initialized 39,264-parameter, two-layer, three-head Transformers from scratch on all 100 ordered pairs of addition modulo 10. Their frozen vocabulary has digits, a plus token, and an equals token; every seed reached 100% training accuracy. This deliberately small, in-sample task is not a benchmark of mathematical reasoning or generalization.

For each clean prompt $(a, +, b, =)$, we corrupted the left addend with $a + 1 \pmod{10}$. At each layer and position, we replaced the corrupted post-block residual vector with the clean vector and measured recovery of the clean answer logit relative to clean and corrupted baselines. This is an activation-patching intervention [10, 34]. For prompt i, recovery is

$$R_i = \frac{\ell_y(\text{patched}, i) - \ell_y(\text{corrupt}, i)}{\ell_y(\text{clean}, i) - \ell_y(\text{corrupt}, i)}.$$

Near-zero denominators are excluded under the fixed protocol. We average R_i within model and use the four initialization seeds as the uncertainty unit, reporting two-sided t intervals with three degrees of freedom.

Layer 1 left-addend patching recovered 0.36 of the clean-logit gap (95% CI [0.20, 0.52]), compared with 0.00 (95% CI [0.00, 0.00]) for the contextual operator control. The preregistered paired difference is 0.36 (95% CI [0.20, 0.52]). Layer 2 equals-position patching recovered 1.00 (95% CI [1.00, 1.00]; Figure 7). The last result is an outcome-position positive control because the unembedding reads that final residual position; it is not evidence for a distributed arithmetic circuit.

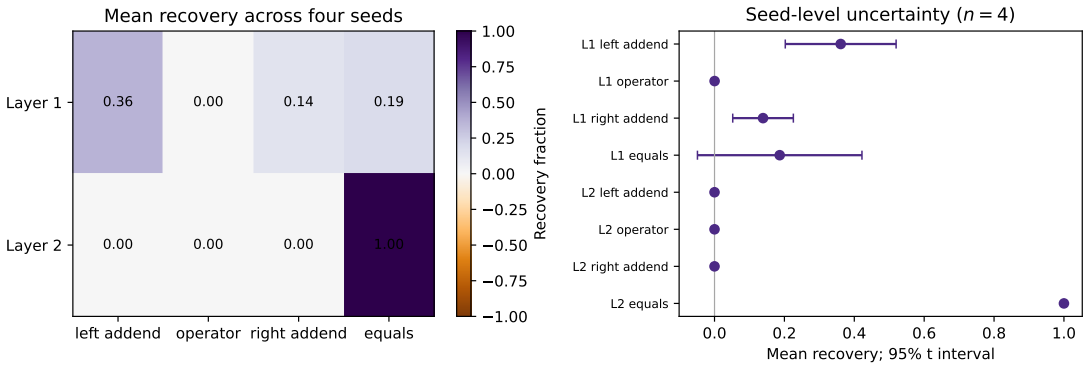


Figure 7: Four-seed causal residual-stream patching in the inspectable toy Transformer. The heat map gives mean clean-answer-logit recovery after inserting a clean residual state into a left-addend-corrupted run; points show the mean and two-sided 95% Student-t interval across four independent initializations. This figure is separate from the OpenAI artifact audit.

The surrogate result demonstrates a repeatable causal-intervention procedure. It does not establish a reusable arithmetic circuit, necessity or sufficiency of any site, generalization beyond its training table, or any property of OpenAI's model.

7 Limitations and research agenda

This audit concerns one release commit; catalogue labels are upstream metadata, not independent adjudication. The exhaustive tree census covers every versioned blob structurally, but it does not read every file semantically. We did not compile the substantial upstream Lean project or assess mathematical validity. PDF extraction is imperfect, although failures are recorded. The toy Transformer was trained on its evaluation table, so its result is methodological rather than evidence of abstract reasoning. Most importantly, public output artifacts cannot identify mechanisms inside an unreleased model.

Version 2 remains partial. Its source-structure outputs meet the corpus integrity and provenance portion of G1 and provide a documented exploratory input for G2, but they have not been independently semantically audited. G3–G7 have not been entered: no valid counterfactual benchmark, behavioral competence result, held-out predictive comparison, causal intervention result, or robustness result exists. CPU-only PyTorch, no cached open-weight model, and no detected Lean toolchain are recorded infrastructure constraints, not negative scientific findings.

A credible next study should choose an open-weight mathematical model, freeze prompts and scoring, collect clean and controlled-corruption runs, and test causal effects through activation patching, path patching, ablation, or feature intervention across seeds and held-out tasks. It should explicitly separate a visible rationale from both the answer and the latent state, and test proposed circuits against multi-site or resampling interventions rather than treating a single patching map as confirmation.

A useful shared next artifact would be a versioned causal-transfer benchmark: a corpus-derived task taxonomy, fixed prompt and counterfactual generator, formal or otherwise transparent scoring suite, model-and-seed matrix, and intervention registry all frozen before outcomes are observed. It should retain only candidate explanations that predict intervention effects across model families and held-out task variants, and document licensing and training-data contamination risks. Such a benchmark would enable task-specific comparisons of sparse probes, sparse feature circuits, and

feature dictionaries under shared control and interpretability criteria [12, 17, 18]. This is a proposed research infrastructure, not a claim that the current audit has identified an OpenAI circuit.

8 Conclusion

The pinned 2026 mathematics release is not a window into an unreleased model's circuitry. It is, however, a durable public specification of a demanding mathematical-output domain: 372 result families, 719 linked manuscripts, Lean scope materials, and 10 released summaries, now situated within an exhaustive 133,955-blob release-tree census. This study releases a pinned, auditable map of that domain together with a deliberately separate causal-intervention template.

That division makes the work actionable. Future studies can preregister behavioral and structural hypotheses from the public record; instantiate matched prompts, perturbations, and scoring rules in inspectable open-weight mathematical models; and retain only candidate explanations that predict intervention effects across seeds, model families, and held-out task variants. Such studies would not identify OpenAI's mechanism. They could establish the independently valuable result that particular causal computational strategies are sufficient and robust within the tested model population for behaviors the release makes observable.

The contribution is therefore a reproducible bridge from public mathematical artifacts to falsifiable mechanistic-interpretability research: public outputs define the phenomenon to be explained, while interventions in accessible models supply the evidence required to explain it. Keeping those evidence types separate is not a retreat from impact; it is what makes a scalable, cumulative research program possible.

Reproducibility and ethics

Derived outputs are regenerated by the analysis, surrogate, and Version 2 proof-extraction modules documented in the repository README. The source archive excludes third-party corpus files. The full-tree census is specified in the accompanying inventory protocol. This study makes no claim about proprietary model internals, intent, or hidden reasoning. The complete research repository is available at the project repository URL.

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